

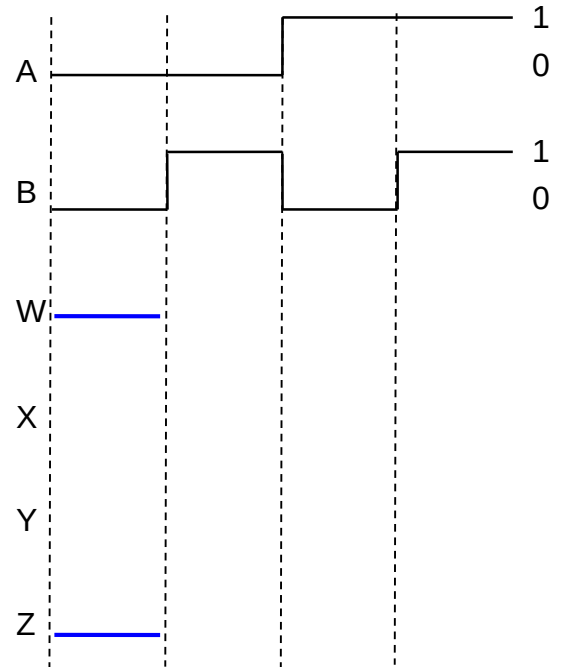
Tutorial 7 Questions (Part 2)

Introduction to Verilog

Please try out these self-study questions (labeled “SS”). These will not be discussed in class and solutions will be provided later.

- SS1. For the Verilog program below, complete the timing diagram shown for signals **W**, **X**, **Y** & **Z** given that the input signals **A** & **B** are as shown.

```
module q1_v (input A, B, output reg W, Z);
    reg X, Y;
    always @ ( B )
    begin
        W <= A ^ B;
        X = ~ W; //old or new value of W?
        Y = A & B;
        Z <= Y; //old or new value of Y?
    end
endmodule
```



Verilog Modeling

Check that the syntax of your programs is correct by using the Xilinx tool!

- Data encryption schemes are commonly used to secure data privacy. Write a Verilog program for a *digital decrypter* which has a 5-bit input, DATA_IN of the form “C₂O₁C₁O₀C₀” where the operation applied to the three bit data “C₂C₁C₀” to produce the 3-bit output DATA_OUT depends on the two bits “O₁O₀” as shown in the table below.

“O ₁ O ₀ ”	DATA_OUT
“00”	“C ₂ C ₁ C ₀ ”
“01”	“C ₀ C ₂ C ₁ ”
“10”	“C ₁ C ₀ C ₂ ”
“11”	“ $\overline{C_2}\overline{C_1}\overline{C_0}$ ”

2. A multiplexer needs to select one out of five 4-bit input words **A**, **B**, **C**, **D** or **E**, and output it to the 4-bit output signal **T**. The selection must be done according to the selection control inputs S_0 – S_2 (see table below). Write a Verilog program for the multiplexer.

$S_2S_1S_0$	000	001	010	011	100	101	110	111
T	A	B	A	C	A	D	A	E

3. A machine receives two 3-bit unsigned inputs **X**, **Y** and outputs a 6-bit unsigned output **Z** where **Z** is the product of **X** and **Y**. Write a Verilog program that implements this machine without using the multiplication operator.

Hint: Use the shift and add method to perform multiplication i.e.

[illegible]

Note that the addition operation can be performed as follows:

$c \leq a + b$; where a, b & c are standard logic vectors of the same length.

4. The 74'74B is an integrated circuit containing negative-edge triggered D flip-flops with synchronous set and asynchronous reset inputs. The D flip-flop receives four 1-bit input signals, **D**, **CLK**, **S**, **CLR** and produces two 1-bit output signals, **Q** and **QB**. Write a Verilog program that implements the D flip-flop according to the characteristic table shown below.

CLK	CLR	S	D	Q ⁺
X	0	X	X	0
↓	1	0	X	1
↓	1	1	0	0
↓	1	1	1	1

